

PURVA TAGDE  
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BATCH B1

```
JavaScript > <> pract4_final_task.html > html > body > script > getThickerBook
1  <!DOCTYPE html>
2  <html>
3  <head>
4  | <title>Smart Library Management System</title>
5  </head>
6  <body>
7  <script>
8
9  function getThickerBook(a, b) {
10 |   return a > b ? a : b;
11 }
12 function bookFactorial(n) {
13 |   if (n < 0) throw new Error("Negative not allowed");
14 |   let fact = 1;
15 |   for (let i = 1; i <= n; i++) fact *= i;
16 |   return fact;
17 }
18 const checkLibraryNote = (note) => note.toLowerCase().includes("library");
19 function secretShelf(pass) {
20 |   const note = "Rare Book Note: Ancient Manuscript";
21 |   return function(tryPass) {
22 |     return tryPass === pass ? note : "Access Denied";
23 |   };
24 }
25 var libraryName = "City Central Library";
26 function scopeDemo() {
27 |   let totalBooks = 5000;
28 |   if (true) {
29 |     const borrowed = 20;
30 |     console.log("Inside block: Borrowed =", borrowed);
31 |   }
32 |   console.log("Inside function: Total Books =", totalBooks);
33 }
34 console.log("Global: Library Name =", libraryName);
35
36 function reverseISBN(num) {
37 |   if (isNaN(num)) throw new Error("Invalid ISBN");
38 |   let rev = parseInt(Math.abs(num).toString().split("").reverse().join(""));
39 |   return num < 0 ? -rev : rev;
40 }
41 function isBookIdPalindrome(id) {
42 |   let str = id.toString();
43 |   return str === str.split("").reverse().join("");
44 }
45 function isTitlePalindrome(title) {
46 |   let clean = title.toLowerCase().replace(/[^\w-0-9]/g, "");
47 |   return clean === clean.split("").reverse().join("");
48 }
49 function sqrtPages(p) {
50 |   if (isNaN(p)) throw new Error("Invalid input");
51 |   if (p < 0) throw new Error("Negative not allowed");
52 |   return Math.sqrt(p);
53 }
```

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53 }
54 function isPrimeShelf(n) {
55     if (isNaN(n) || n < 2) throw new Error("Invalid number");
56     for (let i = 2; i <= Math.sqrt(n); i++) {
57         if (n % i === 0) return false;
58     }
59     return true;
60 }
61
62 while (true) {
63     let choice = prompt(
64         "Enter your choice (1-8):"
65     );
66
67     if (choice === null || choice === "9") {
68         console.log("Exiting");
69         break;
70     }
71
72     try {
73         switch (choice) {
74             case "1":
75                 let a = parseInt(prompt("Enter pages of Book 1:"));
76                 let b = parseInt(prompt("Enter pages of Book 2:"));
77                 console.log("Thicker book has pages:", getThickerBook(a, b));

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77         console.log("Thicker book has pages:", getThickerBook(a, b));
78         break;
79         case "2":
80             let n = parseInt(prompt("Enter a number for factorial:"));
81             console.log("Library ticket ID (Factorial):", bookFactorial(n));
82             break;
83         case "3":
84             let note = prompt("Enter a note:");
85             console.log("Contains a 'Library'?", checkLibraryNote(note));
86             break;
87         case "4":
88             let pass = prompt("Set a password for rare shelf:");
89             let shelf = secretShelf(pass);
90             let attempt = prompt("Enter password to retrieve note:");
91             console.log("rare book note access?:", shelf(attempt));
92             break;
93         case "5":
94             let isbn = parseInt(prompt("Enter ISBN number:"));
95             console.log("Reversed ISBN:", reverseISBN(isbn));
96             console.log("Is ISBN number palindrome?", isBookIdPalindrome(isbn));
97             break;
98         case "6":
99             let title = prompt("Enter a book title:");
100             console.log("Is book title palindrome?", isTitlePalindrome(title));
101             break;
102         case "7":

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102     case "7":
103         let p = parseInt(prompt("Enter total pages read:"));
104         console.log("Square root of total pages read:", sqrtPages(p));
105         break;
106     case "8":
107         let shelfNum = parseInt(prompt("Enter shelf number:"));
108         console.log("Is Shelf number prime?", isPrimeShelf(shelfNum));
109         break;
110     default:
111         console.log("Invalid");
112 }
113 } catch (err) {
114     console.log("Error:", err.message);
115 }
116 }
117 scopeDemo();
118 </script>
119 </body>
120 </html>

```

**OUTPUT :**

**CASE 1: Compare two books and return the one with more pages.**

127.0.0.1:5500 says

Enter your choice (1-8):

OK

Cancel

127.0.0.1:5500 says

Enter pages of Book 1:

OK

Cancel

127.0.0.1:5500 says

Enter pages of Book 2:

OK

Cancel

```
Global: Library Name = Smart Library      pract4_final_task.html:34
Thicker book has pages: 400              pract4_final_task.html:77
Exiting                                   pract4_final_task.html:68
Inside block: Borrowed = 20              pract4_final_task.html:30
Inside function: Total Books = 5000      pract4_final_task.html:32
```

### CASE 2: Generate a factorial-based library ticket ID.

### CASE 3: Check if a note contains the word “Library”.

### CASE 4: Store and retrieve a rare book note using a closure.

127.0.0.1:5500 says  
Enter your choice (1-8):  
  

OK Cancel

127.0.0.1:5500 says  
Set a password for rare shelf:  
  

OK Cancel

127.0.0.1:5500 says  
Enter password to retrieve note:  
  

OK Cancel

Console AI assistance What's new

top Filter Default levels No Issues

Global: Library Name = Smart Library [pract4\\_final\\_task.html:34](#)

rare book note access?: Rare Book Note: Ancient Manuscript [pract4\\_final\\_task.html:91](#)

Exiting [pract4\\_final\\_task.html:68](#)

Inside block: Borrowed = 20 [pract4\\_final\\_task.html:30](#)

Inside function: Total Books = 5000 [pract4\\_final\\_task.html:32](#)

## CASE 5: Reverse an ISBN number and check if it's a palindrome.

127.0.0.1:5500 says  
Enter your choice (1-8):  
  

OK Cancel

127.0.0.1:5500 says  
Enter ISBN number:  
  

OK Cancel

Console AI assistance What's new

top Filter Default levels No Issues

Global: Library Name = Smart Library [pract4\\_final\\_task.html:34](#)

Reversed ISBN: 4554 [pract4\\_final\\_task.html:95](#)

Is ISBN number palindrome? true [pract4\\_final\\_task.html:96](#)

Exiting [pract4\\_final\\_task.html:68](#)

Inside block: Borrowed = 20 [pract4\\_final\\_task.html:30](#)

Inside function: Total Books = 5000 [pract4\\_final\\_task.html:32](#)

### CASE 6: Check if a book title is a palindrome.

127.0.0.1:5500 says

Enter your choice (1-8):

127.0.0.1:5500 says

Enter a book title:

OK Cancel

OK Cancel

Console AI assistance What's new

top Filter Default levels No Issues

Global: Library Name = Smart Library [pract4\\_final\\_task.html:34](#)

Is book title palindrome? true [pract4\\_final\\_task.html:100](#)

Exiting [pract4\\_final\\_task.html:68](#)

Inside block: Borrowed = 20 [pract4\\_final\\_task.html:30](#)

Inside function: Total Books = 5000 [pract4\\_final\\_task.html:32](#)

### CASE 7: Find the square root of total pages read with error handling.

127.0.0.1:5500 says

Enter your choice (1-8):

127.0.0.1:5500 says

Enter total pages read:

OK Cancel

OK Cancel

Console AI assistance What's new

top Filter Default levels No Issues

Global: Library Name = Smart Library [pract4\\_final\\_task.html:34](#)

Square root of total pages read: 7 [pract4\\_final\\_task.html:104](#)

Exiting [pract4\\_final\\_task.html:68](#)

Inside block: Borrowed = 20 [pract4\\_final\\_task.html:30](#)

Inside function: Total Books = 5000 [pract4\\_final\\_task.html:32](#)

### CASE 8: Check if a shelf number is prime.

127.0.0.1:5500 says

Enter your choice (1-8):

8|

OK

Cancel

127.0.0.1:5500 says

Enter shelf number:

17|

OK

Cancel

⋮

Console

AI assistance

What's new

✕

🔍

🚫

top ▼

👁

Filter

Default levels ▼

No Issues

⚙

Global: Library Name = Smart Library

[pract4\\_final\\_task.html:34](#)

Is Shelf number prime? true

[pract4\\_final\\_task.html:108](#)

Exiting

[pract4\\_final\\_task.html:68](#)

Inside block: Borrowed = 20

[pract4\\_final\\_task.html:30](#)

Inside function: Total Books = 5000

[pract4\\_final\\_task.html:32](#)

>

## ALL CASES IN ONE:

⋮

Console

AI assistance

What's new

✕

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🚫

top ▼

👁

Filter

Default levels ▼

No Issues

⚙

Global: Library Name = Smart Library

[pract4\\_final\\_task.html:34](#)

Thicker book has pages: 540

[pract4\\_final\\_task.html:77](#)

Library ticket ID (Factorial): 120

[pract4\\_final\\_task.html:81](#)

Contains a 'Library'? true

[pract4\\_final\\_task.html:85](#)

rare book note access?: Rare Book Note: Ancient Manuscript

[pract4\\_final\\_task.html:91](#)

Reversed ISBN: 4554

[pract4\\_final\\_task.html:95](#)

Is ISBN number palindrome? true

[pract4\\_final\\_task.html:96](#)

Is book title palindrome? true

[pract4\\_final\\_task.html:100](#)

Square root of total pages read: 7

[pract4\\_final\\_task.html:104](#)

Is Shelf number prime? true

[pract4\\_final\\_task.html:108](#)

Exiting

[pract4\\_final\\_task.html:68](#)

Inside block: Borrowed = 20

[pract4\\_final\\_task.html:30](#)

Inside function: Total Books = 5000

[pract4\\_final\\_task.html:32](#)

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